- know the effects of protein deficiency in children.
- gain an understanding of lipids as a concentrated source of energy.
- differentiate between healthy and unhealthy fats in the diet.

10.1 Origin and composition of proteins:

Origin

Amino acids are small units that combine to form a protein molecule. Plants synthesise amino acids with the help of bacteria and fungi from : (i) soil, which supplies the necessary nitrogen and sulphur; (ii) water, which provides oxygen and hydrogen; and (iii) atmospheric carbon dioxide, which supplies carbon and oxygen. Animals cannot synthesise amino acids from basic elements, but derive them from ingested plants. Thus, the primary source of all proteins is the vegetable kingdom.

Composition

Proteins are chemical compounds that contain the same atoms as carbohydrate and lipid – Carbon(C), Hydrogen (H) and Oxygen (O). Which is not present in CHO and lipibs. They also contain nitrogen (N) atoms. These Nitrogen atoms give the name 'amino' (nitrogen containing) to the amino acids that are the links in the chains referred to as proteins.

DO YOU KNOW...?

- Proteins are the building blocks of life
- Proteins are used for tissue repair and healing

10.2 Structure of proteins

About 20 different amino acids may appear in proteins. All amino acids share a common chemical 'backbone' and it is these backbones that are linked together to form proteins. Each amino acid also carries a side chain, which varies from one amino acid to another. The side chains make the amino acids differ in sizes, shape and electrical charge. The side chains on amino acids are what makes proteins so varied in comparison with either carbohydrate (or) lipids.

Each amino acid contains a carboxyl (COOH) or acidic group and an amino (NH_2) or basic group. The amino acids are mostly linked together in forming a protein molecule through NH_2 group of one amino acid condensing with COOH group of another amino acid with the elimination of one molecule of water, and a compound thus formed is called a peptide and the linkage is called '*peptide linkage*'.

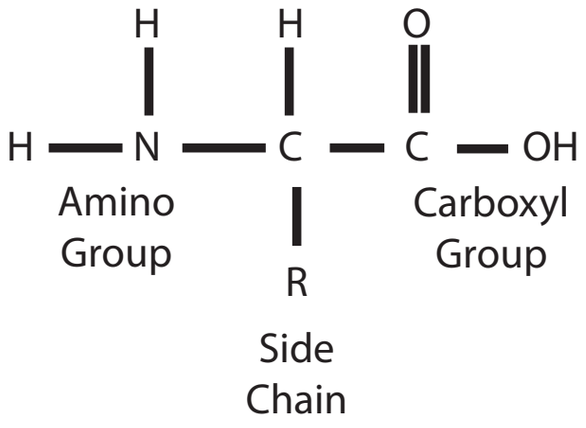


Fig 10.1: Amino acid structure

Protein chains

The 20 amino acids can be linked end-to-end in a virtually infinite variety of sequences to form proteins. When two amino acids bond together, the resulting

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